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ETH AI CENTER

A data flywheel for public, sovereign LLM platforms

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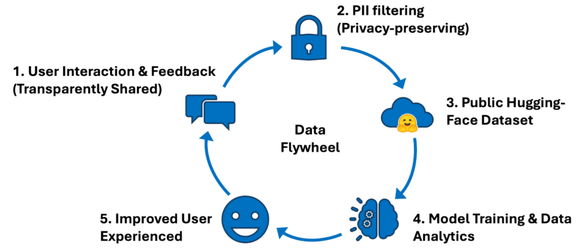


Figure 1: Flywheel Diagram. Data contributions are used for analytics and model training to continuously improve the LLM platform, thereby increasing the volume of users and contributions.

1. Problem Statement

Apertus (Hernández-Cano et al., 2025) is a Swiss sovereign large language model (LLM), hosted on the *Public AI Inference Utility* (Metagov, 2025), designed to align with Swiss values and public access.

Improving the LLM and designing tools to benefit the Swiss user-base requires collecting user interaction data, including **chatlogs and feedback**, but must **ensure strict privacy and user control**. Ethical data collection enables a data flywheel: user interactions create data that improves the model and platform, thereby encouraging further use.

Our work proposes a transparent, privacy-preserving data contribution pipeline tailored to sovereign LLM platforms.

2. Related work

- Industry platforms such as *ChatGPT* use reinforcement learning with human feedback (RLHF) to improve models based on human preferences (L. Ouyang et al., 2022), **but often lack transparency in data collection practices** and do not publish datasets (Alzamil et al., 2025).
- The open-source *Open WebUI* platform supports thumbs up/down feedback collection, but provides no mechanism for sharing contributions to usable public datasets (Baek, 2024).
- LLM training typically relies heavily on web-scraped datasets, raising ongoing concerns around data provenance, licensing, transparency, and ethical use (N. Kandpal, B. Lester, C. Raffel et al., 2025).

3. Data

We collect **opt-in chatlog contributions** from users of the Public AI Inference Utility. A contribution includes a sanitized chat transcript, structured metadata, and explicit usage permissions selected by the user.

Each component of the contribution is controlled by the user, and has schemas to produce an analysis-ready dataset.

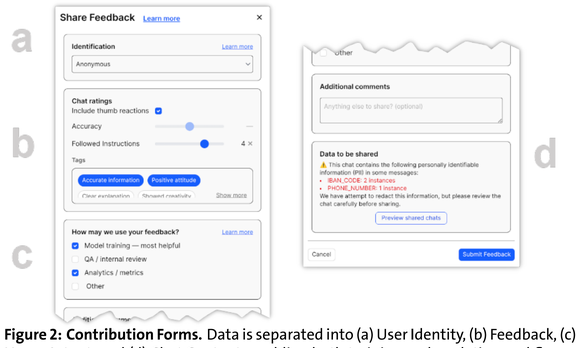


Figure 2: Contribution Forms. Data is separated into (a) User Identity, (b) Feedback, (c) Usage Intent, and (d) Chat Content, enabling both training and analytics workflows.

4. Methods

DATA COLLECTION. We extend the Open WebUI platform to enable structured, privacy-preserving data contributions:

- **Backend:** Performs data sanitization and submits pull requests to a HuggingFace dataset repository.
- **Frontend:** Collects user privacy choices and displays data prior to submission.

The collection process is modular, allowing sovereign platforms to customize consent types, sanitization rules, and dataset schema.

PRIVACY PRESERVATION. Multiple mechanisms ensure user data agency:

- **PII stripping:** Personally identifiable information (PII) is automatically redacted using Presidio, and can be reviewed by users before submission.
- **User data control:** Users control which data elements to include or exclude from their chatlog contributions, such as attribution, thumbs up/down, ratings, or comments.
- **Data usage intent:** Users explicitly specify how their data may be used – for model training, usage analytics, or quality assurance.

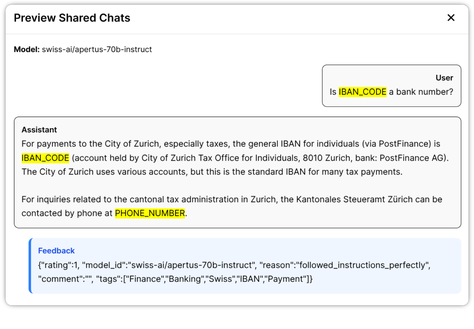


Figure 3: Data Preview. Before submission, the user is shown all data that will be shared. Personal information is redacted automatically, and feedback is only included when specified.

5. Results

We successfully deployed a transparent, privacy-preserving data contribution pipeline including: a **user-friendly UI** to initiate and review data contributions; automatic notification and **redaction of personal information**; consistently structured additions to a **public dataset** for downstream use-cases; and full respect for **user-defined privacy and data usage preferences**.

FLYWHEEL ENABLEMENT. Our research uncovered diverse applications beyond RLHF which utilize ethically-collected chatlog data:

- *Text mining:* Researchers, policymakers, and entrepreneurs can analyze population-scale chats to identify emerging topics and common user needs.
- *Tool development:* Developers can study real user interactions to design new capabilities and improve failure modes of existing tools.
- *Public training data:* Contributions can support training and evaluation not only for Apertus but also for other public-domain LLMs.

6. Conclusions

Ethical data collection on LLM platforms is technically feasible and can sustain a privacy-preserving data flywheel.

Our work shows that users can retain control over their data through granular consent mechanisms and data previews. The collected data enables both traditional RLHF training and novel analytics applications, such as text mining over population-scale chatlogs.

7. Future Work

- Develop MCP-based data collection methods to reduce vendor lock-in and enable contextual prompts encouraging users to contribute.
- Build public frameworks for analyzing collected data, enabling an open analytics ecosystem and closing the data flywheel loop.

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Contributions

Author contributions using the CRediT framework:
PA & ED: Methodology, Software, Visualization, Investigation, Writing
JL & JT: Conceptualization, Resources, Supervision
IS: Supervision, Project Administration

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